

# Kaustav Mukherjee

415-216-5804 | [kaustav.mu@gmail.com](mailto:kaustav.mu@gmail.com) | [linkedin.com/in/kaustavmu](https://linkedin.com/in/kaustavmu) | [github.com/kaustavmu](https://github.com/kaustavmu) | [kaustavmukherjee.com](https://kaustavmukherjee.com)

## EDUCATION

|                                                                                                                           |                                              |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Carnegie Mellon University (CMU)</b><br><i>Master of Science in Computer Vision (GPA 4.22/4.33)</i>                    | Pittsburgh, PA<br>Aug. 2024 – Dec. 2025      |
| <b>National University of Singapore (NUS)</b><br><i>Bachelor of Engineering in Mechanical Engineering (GPA 4.64/5.00)</i> | Singapore, Singapore<br>Aug. 2020 – May 2024 |

## EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>AI Engineer</b><br><i>Kaliber Labs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Jan. 2026 – Present<br>San Francisco, CA   |
| <ul style="list-style-type: none"><li>Developed a training-time, fourier-symmetrized auxiliary objective for JEPA-style world models to enforce physically-informed structuring of the latent space, improving baselines by 50% in low-data scenarios</li><li>Built a real-time multi-view 3D reconstruction pipeline from surgical video using TSDF fusion, driving a 2-day showcase resulting in a signed LOI with a Fortune 500 medical device company and a paper accepted at ECCV</li><li>Improved textual and aural emotion recognition models by architecting a multi-task learning (MTL) framework for categorical classification and regression, driving a 30% absolute jump in accuracy</li></ul>                   |                                            |
| <b>AI Engineer Intern</b><br><i>Kaliber Labs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | May 2025 – Aug. 2025<br>San Francisco, CA  |
| <ul style="list-style-type: none"><li>Architected a modular robotic navigation system with ROS2 and Nav 2, integrating 6+ CV models with C++ behavior trees via shared memory and multithreading, and deployed on the LG Q9 robot and NVIDIA Isaac Sim</li><li>Reduced per-frame edge processing latency from 300ms to 150ms across multiple CV inference tasks</li><li>Led technical integration across 10+ team members and delivered 10+ live demos to investors and stakeholders</li></ul>                                                                                                                                                                                                                                |                                            |
| <b>Software Engineering Intern</b><br><i>Kaliber Labs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Jan. 2023 – Jul. 2024<br>San Francisco, CA |
| <ul style="list-style-type: none"><li>Accelerated convergence of a surgical camera pose estimation algorithm by 10 times using particle swarm optimization (PSO) to match semantic segmentation masks between the scope view and a digital twin</li><li>Utilized NeRFs for real-time camera registration, enabling geodesic measurements with 90% accuracy</li><li>Fine-tuned stable diffusion to generate synthetic training data, expanding surgical tool datasets by up to 5 times</li><li>Employed artificial data augmentation to train multiple semantic segmentation models to above 97% accuracy</li><li>Optimized CV model inference pipelines via ONNX and quantization, reducing memory footprint by 40%</li></ul> |                                            |

## PUBLICATIONS

Bychkov, D., Shah, R., Mukherjee, K., So, M. "Surface Coverage Completeness for Quality-Assured Arthroscopic 3D Reconstruction" *Proceedings of the ECCV Workshop on DCA-MI, 2026*

Mukherjee, K. et al. "Geometry-Conditioned Latent World Models with Recurrent Adaptation in Dynamic Environments" *Submitted to ICLR 2027 (Under Review)*

Zhu, P., Penachio, S., Mukherjee, K., Jonelagadda, A. "Spectral-Target Physical Latent Structuring for JEPA-Style World Models" *Submitted to ICLR 2027 (Under Review)*

## PROJECTS

|                                                                                                                                                                                                                                             |                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Bundle Adjustment for 360° Pose Correction</b>   <i>Gaussian Splatting, SfM, SLAM</i>                                                                                                                                                    | Aug. 2025 – Nov. 2025 |
| <ul style="list-style-type: none"><li>Implemented a rig-based Structure from Motion (SfM) pipeline with bundle adjustment for Stella-VSLAM outputs from 360° images, improving Gaussian Splatting reconstruction quality by 10.5%</li></ul> |                       |
| <b>Drone Perception System at CMU Airlab</b>   <i>Python, OpenCV, ROS, Computer Vision</i>                                                                                                                                                  | Aug. 2024 – Nov. 2024 |
| <ul style="list-style-type: none"><li>Designed a depth-aware perception system for a drone with a 2-DOF manipulator using ROS and RGBD cameras</li></ul>                                                                                    |                       |

## SKILLS

**Competencies:** NeRF, Gaussian Splatting, World Models, 3D Reconstruction, Computer Vision, SLAM, SfM, Diffusion, Reinforcement Learning, LLM Fine-tuning (RLHF, DPO), Context Engineering, Model Quantization  
**Frameworks:** PyTorch, Tensorflow, NumPy, OpenCV, ROS2  
**Languages and Tools:** Python, C++, JavaScript, Docker, Redis, NATS, ONNX, AWS, Linux, Git